

The SAFE Image Authenticity Challenge: Detecting and Localizing Partial and Fully Synthetic Manipulations

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Abstract

Advances in generative AI and AI-assisted image editing tools have made it increasingly easy to produce not only fully AI-generated images, but also more subtle, locally manipulated partially synthetic content that is difficult for both humans and automated systems to detect. This emerging class of mixed-authenticity imagery challenges existing multimedia forensic benchmarks, which typically evaluate only a single manipulation type. Consequently, state-of-the-art forensic methods remain limited in their ability to operate reliably across a diverse range of fake image types and to provide insights beyond binary real-versus-fake decisions. To address this gap, we introduce the SAFE: Image Editing Detection and Localization Challenge, a new benchmark and competition designed to evaluate detection, classification, and localization across a broad spectrum of fake and manipulated images, including fully AI-generated and AI-edited content. Participating systems must output a global authenticity decision, a manipulation type classification, and a pixel-level localization mask. We support SAFE with SAFE-FORGE, a large-scale dataset generated by a scalable forensic data engine that produces diverse fake images with comprehensive ground-truth annotations and traceable manipulation histories. Preliminary evaluations using state-of-the-art forensic systems reveal substantial vulnerabilities in both detection and localization, underscoring the need for unified forensic approaches that remain robust to the rapidly evolving capabilities of modern generative and image editing models.

1. Introduction

Fake and manipulated images pose a growing threat to information security and societal trust. Inauthentic images can be used to facilitate false and misleading in-

formation campaigns, enable fraud and other economic crimes, support identity theft and impersonation, undermine public trust in institutions, and distort public discourse [8, 21, 28, 36]. As images continue to serve as a primary medium for communication and evidence in news, social media, and official reporting, the ability to reliably authenticate visual content has become increasingly critical [2, 27, 79].

Image falsification can occur through a wide variety of mechanisms, spanning both traditional image editing pipelines and modern generative AI systems. For decades, image editing tools such as Adobe Photoshop have enabled users to create visually convincing forgeries by manipulating specific regions of an image or splicing content from other images. These tools contain features that enable a wide variety of image manipulations including object re-sizing and rotation, sharpening, blurring, color and lighting adjustment, content splicing, and object removal.

In parallel, generative AI models for image synthesis have advanced rapidly over the past decade, evolving from early generative adversarial networks (GANs) [41, 43, 44] to diffusion models [24, 70, 75] and, more recently, diffusion-transformer-based architectures [69]. Powerful image generators, such as Stable Diffusion, DALL-E, and Midjourney, are now widely available and accessible to the general public. These models enable the creation of high-resolution, photorealistic images directly from text prompts.

Most recently, generative AI systems have emerged that enable users not only to create images from scratch, but also to edit existing images in a highly flexible and intuitive manner [3, 37, 97]. These models allow targeted image modifications using natural language prompts, optionally guided by spatial masks that localize the regions to be altered. Widely available consumer tools, including Adobe Firefly, Canva, and similar platforms, further lower the barrier to producing subtle, localized manipulations such as object removal, synthesis, and modification.

To address the threats posed by fake and manipulated images, the media forensics community has developed a broad range of image authentication techniques. These include methods for detecting image splicing and localized editing, which aim to identify manipulations produced using traditional image editing tools [35, 49, 62], as well as approaches designed to detect AI-generated synthetic images [64, 66, 93]. Collectively, these methods have demonstrated strong performance on controlled benchmarks and curated academic datasets, providing important progress toward automated image authentication.

However, despite these advances, significant challenges remain in deploying image authentication systems in real-world settings. A key limitation of many existing approaches is that they implicitly assume prior knowledge of the type of manipulation under consideration [71, 90, 91]. For example, detectors are typically designed under the assumption that an image is either fully synthetic, or that it may contain manipulations produced by conventional editing tools such as splicing or retouching, while excluding other possibilities. As a result, these methods are rarely evaluated against a broad and diverse range of forgery types within a unified framework.

Recent research has shown that this assumption leads to critical failure modes. In practice, images may be fully real, fully synthetic (fully AI-generated), or partially synthetic (AI-edited), where an otherwise authentic image contains only a localized region generated or modified using AI. In such cases, both synthetic image detectors and traditional image editing or splicing detectors struggle to reliably identify inauthentic content. Importantly, even when multiple detectors are combined, performance degrades substantially for partially synthetic images containing localized AI-generated content. At present, there are no widely adopted techniques capable of robustly detecting and localizing such manipulations, as most synthetic image detectors are designed specifically to distinguish between fully real and fully synthetic images.

To address this gap, we have organized the **SAFE: Image Editing Detection and Localization Challenge**. This competition challenges participants to identify inauthentic images across a broad variety of forgery types, including content splicing and localized editing, fully AI-generated images, and localized manipulations produced using advanced AI-based image editing tools. Participants are tasked with (i) distinguishing unaltered images from manipulated or AI-generated images, (ii) identifying the type of forgery used to create inauthentic images, and (iii) localizing manipulated regions in the case of locally edited or AI-edited images. The remainder of this paper describes the SAFE Image Edit and Localization Challenge in detail, and presents the dataset and evaluation protocols used in the competition.

2. Background & Related Work

Categorization of fake and manipulated images. Modern image synthesis and editing tools allow anyone to produce fake and manipulated images with ease. These inauthentic images can be broadly divided into four major types: (1) **spliced images** where an object from a source image is pasted into a victim image (copy-move is when the source and victim image are the same image) [83, 90, 103]; (2) **traditionally edited images** where a part or the whole image is processed by classical operations such as blurring, Gaussian noise, color correction, or resizing [6, 33, 82]; (3) **synthetic images** where the whole image is generated using generative AI models [13, 64, 66, 86, 93]; and (4) **AI-edited images** where only a localized region is retouched, removed, or changed using modern editing models such as Qwen-Edit or FLUX.1 Kontex [23, 85, 87].

Detecting fake and manipulated images. A wide range of forensic algorithms has been developed to detect manipulated or synthetic images, but nearly all methods target only a single manipulation category. These approaches fall into two main families: forensic classifiers and anomaly detectors. **Forensic classifiers** analyze the entire image to infer a specific forensic attribute, such as source provenance (camera model or generator model) [1, 7, 9, 26, 99], evidence of traditional editing operations (blur, noise, resizing, color correction, etc.) [5, 11, 46, 84], or compression characteristics (JPEG, WebP, etc.) [4, 25, 49, 52, 67]. **Anomaly detectors** instead localize regions whose forensic properties are inconsistent with the rest of the image without identifying their semantic or operational cause. Examples of these algorithms include splicing and copy-move detectors [35, 49, 62]. Both families provide valuable capabilities but operate under narrow assumptions, limiting their generality.

Traditional forensic algorithms. Before the advent of generative AI models such as GANs [40–44] and diffusion models [24, 69, 70, 74, 75], research primarily focused on camera model identification [7, 9, 61, 89], traditional image editing detection [5, 11, 46, 58, 60, 80, 84], and splicing/local editing localization [14, 17, 18, 35, 39, 49, 55, 62, 96]. These algorithms often rely on forensic microstructures - content-independent, pixel-level correlations introduced by the image formation process and any subsequent post-processing operations [18, 35, 60, 62, 64]. When these microstructures are extracted using human-engineered mathematical models [20, 29, 30, 32] or deep neural networks such as CNNs [6, 7, 9, 10, 100, 101], they can be used to infer source, detect editing, assess compression, or reveal regions with inconsistent microstructures [14, 18, 35, 39, 63, 96].

Forensic algorithms addressing generative AI. After generative AI emerged and surged in popularity, researchers developed detectors for fully synthetic images [13, 64, 66, 86,

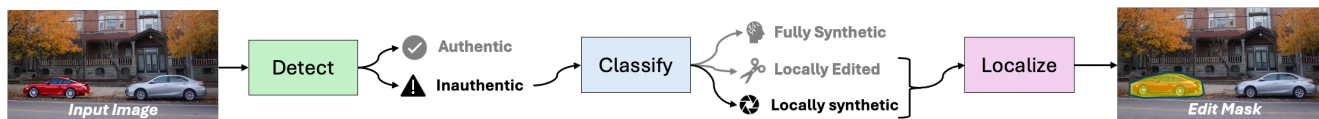


Figure 1. Example showing three output requirements: detection, classification, and localization, in the SAFE challenge.

93], as well as specialized systems for detecting face-swap (more commonly known as deepfake) content [16,57,59,77,81,98,102]. Multiple works demonstrate that forensic microstructures remain highly effective because AI-generated images exhibit formation mechanisms fundamentally distinct from physical cameras [53,57,59,64,65]. Some recent face-swap deepfake detectors combine microstructure cues with semantic anomaly analysis - e.g., detecting unnatural textures, reflections, edges, or anatomical inconsistencies [15,51,54,56,88,95]. However, as generative models continue to rapidly improve, these semantic artifacts diminish, and semantic-based approaches may become less reliable [19,72,73].

Detecting AI-edited images remains a challenge. A newly emerging generative capability is AI-based image editing - AI models that allow users to faithfully change the content of an image by simply providing a prompt with an optional mask specifying the editing region [3,37,50,94,97]. These generative models will often change a localized part of the image while leaving the rest of the image largely untouched. While anomaly-based tools such as splicing detectors can be applied, their performance on modern AI-edited images remains untested and may degrade to the significant difference in forensic microstructures of AI-generated content versus those of real cameras.

Overall, while strong algorithms exist for detecting each individual category of fake images, there remains no unified forensic system capable of detecting all categories without prior knowledge of which type of manipulation occurred. This gap is a fundamental obstacle preventing current forensic approaches from being operationally viable in real-world conditions [31,38,45,71,92].

Previous Synthetic Media Detection Challenges. Researchers have conducted various synthetic media detection challenges in the last several years. Some of the earliest competitions were put on by Meta and academic researchers [22,76,78]. However, these challenges focused on facial manipulations in videos that were prominent at the time. More recent challenges include the SAFE Audio challenge [47] and the SAFE Video challenge [48], but these challenges operate in audio and video modality, respectively. NIST hosted the closest relevant challenge on fully generated and inpainted image detection [34], but participation was limited.

3. Challenge Design and Tasks

The SAFE: Image Editing Detection and Localization Challenge is designed to evaluate the next generation of image forensic methods capable of distinguishing among multiple forms of image manipulation and providing spatially precise localization of altered content. Unlike prior benchmarks and detection tasks that focus primarily on one particular category of fake images, *e.g.* fully AI-generated, or splicing imagery, the SAFE challenge introduces a structured, multi-task evaluation that reflects the complexity and subtlety of modern image manipulation techniques.

At a high level, the challenge focuses on detecting fake and manipulated images arising from four distinct generation or modification processes: (1) **splicing**, where content from another real image is inserted into an image; (2) **traditional editing**, where a portion of the image is altered using conventional editing tools such as Photoshop or standard image processing operations (e.g., blurring, Gaussian noise, or color correction); (3) **fully AI-generated images**, which are synthesized end-to-end using text-to-image or image-to-image generative models; and (4) **AI-edited images**, where localized regions are modified using AI-based image editing models. These categories span the spectrum of fake images, but crucially also include mixed-authenticity cases in which AI-generated (synthetic) materials or materials that were transformed using traditional means are embedded into an otherwise real scene. This framework enables the community to study and benchmark forgeries that represent realistic attack vectors seen in real-world information sharing and contextual changes, commercial editing workflows, and AI-assisted content generation platforms.

Therefore, in this challenge, we introduce three primary evaluation tasks, detection, classification of the fake category, and localization of fake content. First, participants must perform manipulation detection, determining whether an image is authentic or inauthentic by treating the latter three categories as a unified “fake” super-class. This allows models to be assessed on their ability to detect the presence of manipulation regardless of type. Second, participants must perform fine-grained classification, predicting the specific manipulation category for each image. This multi-class evaluation highlights whether models can distinguish between traditional editing, generative insertions, and fully synthetic content, which is an increasingly important distinction for analysts, journalists, and downstream trust

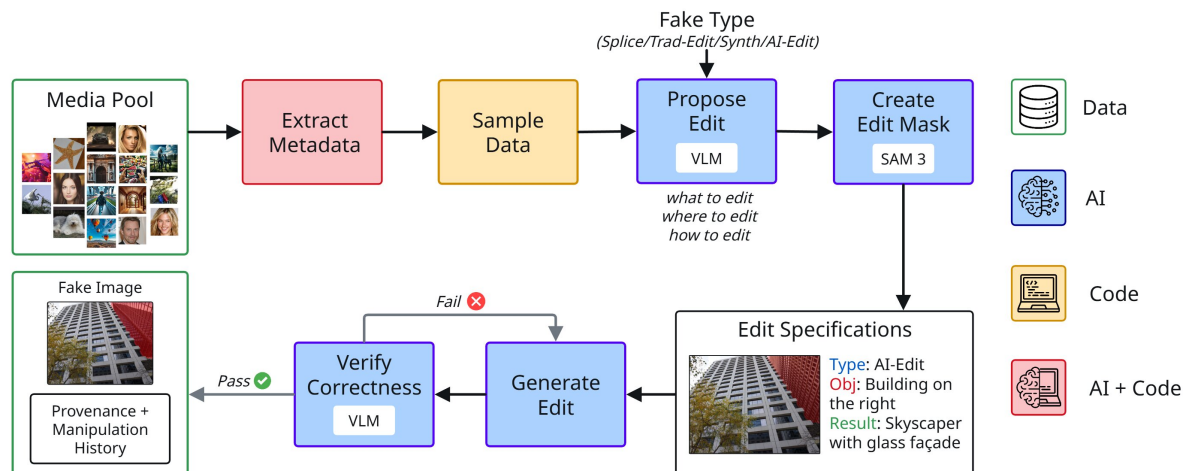


Figure 2. Overview of our scalable forensic data engine that produces SAFE-FORGE - SAFE Forensic Generalization - dataset.

frameworks. Third, the challenge emphasizes localization, requiring participants to produce binary masks that identify manipulated regions. Localization performance is evaluated with established metrics in the field of image forensics, such as the F1 score or Matthews Correlation Coefficient (MCC), enabling robust comparisons across models and manipulation types. This task encourages the development of interpretable systems that provide spatial evidence of manipulation, a key requirement for operational use in digital forensics, media integrity reviews, and human-AI decision support pipelines. An example of such systems is presented in Fig. 1 where an input image is first determined if it is authentic and subsequently identified its fake region if the image is classified to be locally edited.

The SAFE: Image Editing Detection and Localization Challenge is hosted on the Dyff AI evaluation platform, which provides a standardized testing environment, blind test data, and automated scoring pipelines. Participants submit model repositories that are executed within a controlled environment, ensuring fairness, reproducibility, and consistent resource allocation. This infrastructure further supports reproducible research through transparent evaluation protocols and leaderboard reporting that will accompany eventual workshop presentations.

4. Dataset and Data Engine

Overview. In order to remove any confounding factors that may affect the evaluation of image authentication systems and understand their capabilities in a variety of scenarios, we require a comprehensive dataset of authentic and fake images in which their provenance, manipulation history, and ground-truth structure are well-documented and controlled. To this end, we developed **SAFE-FORGE: SAFE Forensic Generalization Dataset**, a dataset con-

structed through a data engine designed to generate four major classes of inauthentic imagery - splicing, traditional editing, fully synthetic generation, and AI-based editing - each paired with its pristine camera-original counterpart along with detailed ground-truth metadata. The data engine automates every stage of this process: proposing edits, generating required masks, synthesizing or manipulating regions, verifying correctness, and emitting structured ground-truth metadata. The entire pipeline ensures that each sample contains complete, machine-verifiable ground truth for source, manipulation, and compression attributes. *We note that the specific AI generators or AI image editing models used to create this dataset are currently kept private in order to organize our competition.* A visual overview of our data engine is shown in Fig. 2. In the following sections, we will discuss each modules of our data engine in more detail.

Media Pool and Metadata Extraction. The engine begins with a media pool of pristine, unedited photographs captured by real cameras. Images are sourced from public photography datasets and internal collections, filtered to ensure they are camera-original and accompanied by reliable EXIF fields describing camera make/model, color pipeline, and geometric characteristics. Each image is processed to extract and standardize its EXIF metadata, along with a detailed caption specifying all scene content, objects, and attributes present in the image. This metadata is crucial for guiding subsequent manipulation proposals and ensuring semantic coherence in edits.

Edit Proposal Generation. At the core of the data engine is a proposal module that, given a target fake-image type, samples a structured set of attributes describing how that fake should be created. This module is driven by predefined sampling procedures to select images or objects needed for manipulation, as well as a vision-language model (VLM)



Figure 3. Examples of manipulated images in our SAFE-FORGE dataset.

that interprets the inputs and fills in required details. Proposals are put together in XML format by the VLM so they can be deterministically parsed and validated. Below we describe the information sampled for each fake-image type.

Splicing Proposals. For spliced images, the proposal specifies:

- Host image URL
- Donor image URL
- Donor object
- Target location of the donor object in the host image

The VLM is prompted with the host and donor images to describe plausible objects and placements consistent with scene geometry.

Traditional Editing Proposals. For traditional operations (blur, noise, color correction, resizing, etc.), the proposal includes:

- Image URL
- Object or region to be edited
- Editing operation type

Semantic regions are chosen by prompting the VLM with text queries about salient objects or areas; operation types are sampled from a predefined library.

Fully Synthetic Proposals. For fully synthetic images, the engine generates:

- Original image URL
- Generation method
- Positive and negative prompts

The prompts are derived from detailed captions of the original real image so that the synthetic variant is semantically aligned with its real counterpart.

AI-Editing Proposals. For AI-based editing, the proposal specifies:

- Image URL
- Editing generative model
- Object/region to be edited
- Positive/negative editing prompts

These prompts describe the intended modification (removal, replacement, content alteration) while maintaining scene coherence.

Mask Generation. For splicing, traditional editing, and AI-editing cases, the proposal must be accompanied by a pixel-level mask specifying the edited region. The mask is generated by feeding the VLM-supplied region description into SAM 3 [12], which produces a segmentation mask



Figure 4. Examples of fully AI-generated (synthetic) images in our SAFE-FORGE dataset.

aligned with the described object or region. We note that fully synthetic images do not require a mask because their entire content originates from a non-camera source.

Edit Execution and Generation. Given the proposal and editing mask, the engine constructs a *complete edit specification* and sends the instruction to an appropriate image-editing or image-generation model. The set of instructions is dependent on the type of fake image it is trying to create, for example:

- **Splicing:** the donor region is extracted from the donor image and composited into the host image at the specified location.
- **Traditional editing:** the specified operation is applied with automatically sampled parameters.
- **Fully synthetic generation:** an AI model generates a new full-frame image conditioned on the prompts derived from the real image.
- **AI editing:** the masked region is modified via an AI inpainting/edit model conditioned on the proposed prompts.

Each output image is saved alongside its proposal metadata and mask. The specific AI generators or AI image editing models used are currently kept private in order to organize our competition.

Automated Verification. The engine performs a verification pass using a VLM to ensure that the generated fake image is consistent with the proposal and specifications. No human intervention or fallback is used at any stage. Regarding criteria for splicing, traditional editing, and AI-editing images, the VLM first checks that the edited region corresponds to the intended object/area, then verifies that the visual effect matches the specified operation or prompt. For fully synthetic images, the VLM checks that the generated content matches the semantic prompts derived from the real image. For each image, the VLM outputs if the verification passes or fails. If verification fails, the proposal is discarded and a new one is sampled automatically.

Ground-Truth Metadata Emission. Finally, the engine emits a structured XML file containing comprehensive ground-truth metadata for each image pair (real and fake). This includes: (1) Source image details (URL, EXIF meta-

data), (2) Manipulation type and parameters, (3) Pixel-level editing masks. This metadata is designed to be machine-verifiable, enabling precise evaluation of image authentication systems against known manipulation histories.

Dataset Composition. The engine is run at scale to produce 10,000 images for each fake-image type - splicing, traditional editing, fully synthetic generation, and AI-based editing - yielding 40,000 manipulated samples paired with their 10,000 pristine originals. Some selected examples of these fake images are shown in Figs. 3 and 4.

5. Metrics

Submissions to the SAFE Image Forensics Challenge are evaluated using multiple complementary metrics, each designed to measure a distinct aspect of image authentication performance. Together, these metrics assess a system’s ability to (i) detect inauthentic images, (ii) identify the type of manipulation used to create an inauthentic image, and (iii) localize manipulated regions in images containing partial or localized forgeries.

Image Authenticity Detection. The first performance aspect evaluates each submission’s ability to distinguish unaltered images from all forms of image manipulation and forgery. The evaluation dataset is partitioned into two high-level classes: unaltered images and inauthentic images. The inauthentic class includes fully AI-generated images, AI edited (partially synthetic) images containing localized AI-generated content, and images manipulated using traditional or AI-based editing tools.

For this task, we report balanced detection accuracy, computed as the average of the per-class accuracies for the unaltered and inauthentic classes. Balanced accuracy is used as the primary evaluation metric to account for potential class imbalance and to reflect real-world deployment scenarios, in which an image authentication system must select a detection threshold without prior knowledge of the image’s provenance or manipulation type. In addition to balanced accuracy, we also report the area under the receiver operating characteristic curve (AUC) as a secondary metric to characterize overall detection performance across thresholds.

Table 1. Detection performance breakdowns by fake image type.

Fake Type	Precision	Recall	F1
Full AI Gen	0.77	0.89	0.82
Splice	0.52	0.54	0.53
Trad. Edit	0.21	0.21	0.21
AI Edit	0.26	0.24	0.25
Overall Accuracy	0.43		

Manipulation Type Classification. The second performance aspect evaluates each submission’s ability to distinguish between different categories of image manipulation. For this task, images in the evaluation set are assigned to one of four mutually exclusive classes: unaltered, fully AI-generated, AI edited (partially synthetic), and locally edited images.

Performance is measured using balanced classification accuracy, computed as the average per-class accuracy across the four classes. This metric assesses a system’s ability not only to identify inauthentic images, but also to correctly determine the type of manipulation responsible for the image’s inauthenticity, without bias toward more prevalent classes.

Localization of Inauthentic Content. The third performance aspect evaluates a system’s ability to localize manipulated or synthetic content within images that are not fully synthetic. For this task, participants submit a binary localization mask for each relevant image, indicating pixels corresponding to inauthentic or manipulated regions. These regions may correspond to localized AI-generated content in AI edited (partially synthetic) images, or to spliced or edited regions in traditionally manipulated images.

Localization performance is measured using the F1 score, computed between the predicted localization mask and a ground-truth mask denoting inauthentic regions. The F1 score balances precision and recall, providing a robust measure of localization accuracy for both small and large manipulated regions.

Overall Ranking. Challenge winners are determined based on each participant’s performance across all three evaluation aspects: image authenticity detection, manipulation type classification, and localization accuracy. By jointly considering these metrics, the challenge emphasizes the development of comprehensive image authentication systems capable of robust detection, characterization, and localization of real-world image forgeries.

6. Preliminary Evaluation

Setup. In this section, we present a preliminary evaluation of state-of-the-art forensic systems on the task of detecting and localizing a wide variety of fake image types. Since there currently exists no single unified system designed to operate on all fake types (e.g. spliced, traditionally edited,

Table 2. Confusion matrix (rows: ground truth, columns: predicted) for binary detection classification (real/fake) performance.

GT \ Pred	Full AI Gen	Splice	Trad. Edit	AI Edit	Real
Full AI Gen	0.89	0.01	0.04	0.03	0.04
Splice	0.11	0.54	0.10	0.12	0.11
Trad. Edit	0.09	0.28	0.21	0.20	0.22
AI Edit	0.04	0.16	0.30	0.24	0.26
Real	0.03	0.05	0.32	0.33	0.27

Table 3. Localization F1 scores by fake image type.

Fake Type	F1
Splice	0.68
Trad. Edit	0.30
AI Edit	0.14

fully AI generated, and AI edited), we combined the output of two state-of-the-art systems: TruFor [35], primarily designed to detect and localize splicing and traditional editing, and Community Forensics [68], designed to detect fully AI generated images. To accomplish this, we first feed an input image to both systems and record their outputs. If any system classifies an image as “fake”, then that image is considered to be fake. The image is only considered authentic if both systems classify it as “real”. Additionally, on images in which only a part of them were modified, either using traditional tools or AI models, we disregard the detection decision of both systems and only measure the correlation between TruFor’s produced localization mask and the ground-truth mask. To conduct the competition, we only apply this procedure to a portion of our full SAFE-FORGE dataset, as described in Sec. 4. The result of this experiment is shown in Tabs. 1 to 3 and discussed below.

Results. The detection performance reported in Tabs. 1 and 2 indicates that the combined system of TruFor [35] and Community Forensics [68] achieves strong performance only when detecting fully AI-generated images, with an accuracy of 0.89 and an F1 score of 0.82. In contrast, the system performs poorly on other types of fake images, including spliced images (0.54 accuracy) as well as images edited using AI-based tools (0.24 accuracy) or traditional editing pipelines (0.21 accuracy). Notably, the detection accuracies for all locally edited image categories fall below 0.5, indicating that the system frequently misclassifies such manipulated images as authentic. These results expose a critical limitation of existing forensic solutions when confronted with more realistic and complex image edits, particularly those produced by modern AI-based image editing models.

The localization performance of the combined system is summarized in Tab. 3. While TruFor [35] is able to accurately localize regions manipulated via splicing tools,

achieving an F1 score of 0.68, its performance degrades substantially for images edited using traditional pipelines (0.30 F1) and further deteriorates for images edited using advanced AI-based image editing models (0.14 F1). Given that TruFor remains one of the state-of-the-art methods for splicing and manipulation localization, these results highlight the inability of current systems to generalize beyond narrowly defined manipulation types and to adapt to increasingly realistic, complex, and evolving image editing techniques.

Taken together, these gaps in both detection and localization performance underscore the urgent need for innovating new forensic approaches that can effectively handle a broad spectrum of fake image types while remaining robust to the rapidly changing capabilities of modern generative and editing models.

7. Discussion

The SAFE Image Authenticity Challenge is designed to catalyze a new wave of research in image forensics by shifting the community beyond binary real-versus-fake paradigms and toward richer, fine-grained, and explainable authenticity assessments. The expected impact of the challenge spans methodological, scientific, and operational dimensions.

Classifying Fake Imagery Type. We anticipate that the challenge will help identify fundamental limitations in current forensic approaches, particularly when confronting subtle, spatially localized, or complex manipulations. Existing detectors often rely on global texture cues or generator-specific artifacts, which degrade significantly on partially synthetic images. By requiring per-category scores and manipulation-type discrimination, the challenge encourages the development of models that reason about semantics, context, and local-global interactions, which are capabilities necessary for robustness in real-world conditions.

Localizing Fake Content. The inclusion of a localization track is expected to advance research on explainability and actionable forensic outputs. Detecting where an edit occurred provides critical interpretability for analysts, journalists, and automated downstream systems. We expect this to spur progress in hybrid detectors that integrate feature attribution, spatial anomaly detection, or segmentation-aware architectures.

Inclusion of AI-edited (partially synthetic) Imagery. Detecting AI-edited (partially synthetic) manipulations remains an underexplored problem in existing benchmarks. As generative models proliferate and become embedded in consumer editing tools, many fake images encountered in the wild may contain a hybrid composition of fake and real content, rather than fully synthetic creations. By providing

a benchmark aligned with this emerging generative capability, the challenge supports the development of methods that generalize not only across fake imagery types, but also robust to the evolving landscape of generative AI. This shift mirrors real operational needs in journalism, law enforcement, national security, and platform integrity.

Reproducibility and Transparent Evaluation. The SAFE challenge is positioned to support reproducibility and transparent evaluation, leveraging the Dyff platform’s infrastructure. By initially limiting access to any of the evaluation data we are discouraging overfitting, and moving towards approaches that are generalizable. Over time, we expect the platform to serve as a foundation for cumulative benchmarking, allowing successive iterations of the challenge to incorporate new manipulation types, unseen data sources, and post-processing variations that reflect the evolving landscape of synthetic media.

Societal Impacts. Beyond methodological advances, the challenge also carries broader societal implications. As hybrid synthetic imagery becomes more accessible through tools like Adobe Firefly, Canva, and similar platforms, establishing verifiable standards for authenticity becomes essential for maintaining public trust. The challenge’s focus on interpretable and localized outputs supports the long-term vision of building forensic systems that can be integrated into newsroom workflows, content generation pipelines, and judicial evidence chains.

8. Conclusion

In this work, we introduced the SAFE Image Editing Detection and Localization Challenge and the accompanying SAFE-FORGE dataset to address a fundamental gap in current image forensics: the lack of unified systems capable of detecting, classifying, and localizing diverse forms of image manipulation without prior knowledge of the manipulation type. Our empirical evaluation of state-of-the-art forensic methods demonstrates that, while existing approaches perform well on narrowly defined manipulation categories, they fail to generalize to more realistic and complex editing scenarios, particularly those enabled by modern AI-based editing models. These findings highlight the limitations of current benchmark designs and underscore the need for new forensic paradigms that move beyond binary authenticity decisions toward robust, generalizable, and interpretable image authentication systems capable of operating in rapidly evolving real-world environments.

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